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Roll No

BE-3001(EC)-CBGS

B.E., IV Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Mathematics - III

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Find the Fourier series to represent the function
 $f(x) = x^2$ in $(-\pi, \pi)$.

b) Expand $f(x) = \pi x - x^2, 0 < x < \pi$ in a half range sine series.

2. a) Find Fourier sine transform of $\frac{e^{-ax}}{x}$.

b) Find the cosine transform of $\frac{1}{x^2 + a^2}$

3. a) Use least square method to fit a straight line to the data

x	1	2	3	4
y	3	7	13	21

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- b) Dot a Poisson distribution to the set of observations.

x	0	1	2	3	4
y	122	60	15	2	1

4. a) Fit a straight line to the data:

x:	1	4	2	3	5
y:	3	1	2	5	4

- b) If on an average one ship in every ten is wrecked. find the probability that out of 5 ships expected to arrive 4 atleast will arrive safely.

5. a) Find the inverse Laplace transforms of the followings:

i) $\frac{1}{9s^2 + 25}$ ii) $\frac{5s-18}{9s^2 + 25}$

b) Evaluate $L^{-1} \left\{ \frac{1}{(s+1)(s-2)} \right\}$.

6. a) Obtain hNf range sine series in $0 < x < 2$.

- b) Find the Fourier transform of

$$f(x) = \begin{cases} x, & |x| \leq a \\ 0, & |x| > a \end{cases}$$

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7. a) Find the Fourier sine transform of

$$f(x) = \begin{cases} \sin x, & 0 < x < a \\ 0, & x > a \end{cases}$$

- b) Find Fourier cosine transform of

$$f(x) = \begin{cases} x, & 0 < x < 1 \\ 2 - x, & 1 < x < 2 \\ 0, & x > 2 \end{cases}$$

8. a) Find mean and variance of the following distribution:

$$\begin{array}{l} x_i = 0 \quad 1 \quad 2 \quad 3 \\ x_i = \frac{1}{8} \quad \frac{3}{8} \quad \frac{3}{8} \quad \frac{1}{8} \end{array}$$

- b) Distinguish between discrete random variable and continuous random variables by giving suitable examples.
